

An Exact Two-Mode Intertwining for $SU(2)$ Heat Transport

A machine-checked operator bridge from Haar convolution to a finite Dobrushin transfer

Lluis Eriksson

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Abstract

We give a machine-checked operator bridge between two independently formalized pieces of two-dimensional Yang–Mills transport. The first is the concrete $SU(2)$ heat operator on normalized Haar probability; the second is an exact two-state Dobrushin transfer whose states contain genuine bundled elements of $SU(2)$. The bridge is not an equality of two rates. We construct a common coefficient space $E = \mathbb{R}^2$, an injective continuous lift $\Phi_\infty : E \rightarrow \{f : SU(2) \rightarrow \mathbb{C}\}$ and a bijective finite lift $\Phi_2 : E \rightarrow \mathbb{R}^2$. For $t > 0$, both actual operators intertwine with the same diagonal map

$$D_t = \text{diag}(1, e^{-3t/4}) : \quad H_t \Phi_\infty = \Phi_\infty D_t, \quad W_t \Phi_2 = \Phi_2 D_t.$$

Moreover Φ_∞ preserves the exact Haar pairing on this two-mode sector, so its injectivity is spectral rather than merely set-theoretic. The continuous proof first establishes that Haar convolution by the infinite heat kernel acts on every Chebyshev character χ_n with Casimir eigenvalue

$$\exp\left(-\frac{tn(n+2)}{4}\right).$$

The infinite identity is obtained by finite character orthogonality and dominated convergence, not postulated. The finite lift sends the fundamental coefficient to the exact nonzero sign vector; the carrier matrices satisfy unitarity and determinant one by their Lean proof fields. The result is an exact invariant restriction and finite conjugacy, not a completed spectral theorem for all of $L^2(SU(2))$ and not a claim that $SU(2)$ is finite. Every load-bearing identity compiles in Lean 4 against pinned revisions.

Artifact statement. The new formal source is `YangMills/OS/SU2HeatIntertwining.lean`. Its concrete dependencies are `YangMills/OS/SU2HeatTransport.lean`, the pinned satellite and `YangMills/OS/SU2TransportWitness.lean`. Exact revisions, byte counts, line-ending hashes, build commands, and runtime information are recorded in the release manifest distributed with this paper.

1 Problem and contribution

For a compact gauge group, heat-kernel convolution is the natural transfer mechanism behind the exact solution of two-dimensional Yang–Mills theory. At the formal level, however, three distinctions matter.

First, an abstract transfer theorem is not yet an inhabited gauge-theoretic endpoint. Second, equality of one scalar decay rate is not an operator map. Third, a numerical certificate for a matrix identity is not a Lean witness inhabiting $SU(2)$. The purpose of the present construction is to close precisely this interface by a commuting diagram.

The contribution is the following chain, checked without a project-local axiom and without assuming any of its conclusions:

- (i) finite heat-kernel partial sums act diagonally on every character already present in the sum;
- (ii) dominated convergence upgrades this to the actual infinite heat kernel;
- (iii) the Casimir ordering $3 \leq n(n+2)$ for $n \geq 1$ identifies χ_1 as the sharp slowest non-vacuum mode;
- (iv) evaluation at the identity proves $\chi_1 \neq 0$ exactly;
- (v) the real span of χ_0, χ_1 is embedded by an exact Haar-isometric lift and is invariant under the actual heat convolution;
- (vi) the same coefficient space maps bijectively to the full finite transfer space, with the fundamental coefficient mapping to the sign vector; and
- (vii) both operators commute with the same diagonal step, yielding an actual restriction and conjugacy theorem in Lean, while the earlier nonzero projected gap endpoint remains available as a corollary.

The novelty is deliberately narrower than the exact area-law results already available in the satellite. We do not rename those results as new. Instead, we add the missing spectral map needed by the parent transport endpoint.

2 Canonical endpoint map

Table 1 records the exact division of labor. This is also the minimum interpretation that connects the continuous heat kernel to the published parent theorem. A rate equality alone is too weak; a full completed $L^2(\text{SU}(2))$ spectral theorem is stronger than necessary. The exact middle object is the invariant two-character sector and its induced finite quotient.

Layer	Exact object	Role
Satellite	normalized Haar probability, Chebyshev characters, finite and infinite class heat kernels, character convolution orthogonality	published analytic input
New heat bridge	<code>heatKernel_character_eigen</code> , Casimir-rate ordering, nonzero fundamental mode	continuous spectral input
New intertwining	injective Haar-isometric lift, bijective finite lift, two commuting operator identities	genuine restriction/conjugacy map
Frozen parent	two bundled $\text{SU}(2)$ holonomies, exact two-state covariance, <code>exact_transport_witness</code>	finite non-vacuous endpoint
Abstract endpoint	abstract uniform-gap theorem	converts exact band decay into a projected transfer norm bound

Table 1: Canonical endpoint map. The two middle rows are the new theorem lane.

The parent endpoint is not merely existential at the carrier level. Its type is a sigma type whose elements contain a finite label, a bundled 2×2 complex matrix, a proof of unitarity, a proof of determinant one, and an equality with one of two explicitly constructed holonomies. Hence every carrier element inhabits $\text{SU}(2)$ in Lean.

3 The concrete heat operator

Let μ_H be normalized Haar probability on $SU(2)$. For functions f, h the satellite convolution convention is

$$(f * h)(g) = \int_{SU(2)} f(x) h(x^{-1}g) d\mu_H(x).$$

Write χ_n for the real-valued Chebyshev character of highest-weight label n , regarded as a complex function, and set

$$a_n(t) = \exp\left(-\frac{t n(n+2)}{4}\right).$$

The class heat kernel is the convergent series

$$K_t(g) = \sum_{n=0}^{\infty} (n+1) a_n(t) \chi_n(g).$$

The factor $n+1$ is the representation dimension, while $n(n+2)/4$ is the $SU(2)$ quadratic Casimir in the satellite's normalization.

The input orthogonality statement is already stronger than a scalar inner product identity. It gives the convolution formula

$$(\chi_n * \chi_m)(g) = \begin{cases} \chi_n(g)/(n+1), & n = m, \\ 0, & n \neq m. \end{cases} \quad (1)$$

This normalization cancels the dimension in K_t exactly.

3.1 Finite partial sums

Let $K_t^{<N} = \sum_{n < N} (n+1) a_n(t) \chi_n$. The first new Lean theorem is the finite identity below.

Theorem 3.1 (finite character eigenidentity). *For $m < N$, $t \in \mathbb{R}$, and $g \in SU(2)$,*

$$(K_t^{<N} * \chi_m)(g) = a_m(t) \chi_m(g).$$

Proof. Finite integrability follows from continuity on the compact group. Linearity of the Haar integral turns convolution with the partial sum into a finite sum of integrals. Equation (1) kills every term except $n = m$; the remaining factor $(m+1)$ cancels its reciprocal. The Lean proof expands the integrand before invoking the satellite theorem, so no desired eigenidentity is passed as a premise. $\square$

The formal statement is:

```
theorem heatKernelPartial_character_eigen_of_lt
  (N : Nat) (t : Real) (m : Nat) (hm : m < N) (g : SU2) :
  su2Convolution (su2HeatKernelPartial N t)
    (su2CharacterChebyshev m) g =
  su2ClassHeatWeight t m * su2CharacterChebyshev m g
```

3.2 Passage to the infinite kernel

The infinite theorem requires more than rewriting a formal series. For fixed m and g , define

$$F_N(x) = K_t^{<N}(x) \chi_m(x^{-1}g), \quad F(x) = K_t(x) \chi_m(x^{-1}g).$$

The satellite supplies pointwise convergence $K_t^{<N}(x) \rightarrow K_t(x)$ for $t > 0$ and a summable heat-kernel majorant. It also supplies $|\chi_m(y)| \leq m + 1$. Thus

$$|F_N(x)| \leq \left(\sum_{n=0}^{\infty} M_t(n) \right) (m + 1),$$

a constant integrable against the probability measure μ_H . Dominated convergence passes the limit through the Haar integral. For all $N \geq m + 1$, Theorem 3.1 makes that integral the same constant; uniqueness of limits completes the proof.

Theorem 3.2 (infinite heat-mode identity). *If $t > 0$, then for every $m \in \mathbb{N}$ and $g \in \mathrm{SU}(2)$,*

$$(K_t * \chi_m)(g) = \exp\left(-\frac{t m(m+2)}{4}\right) \chi_m(g).$$

The crucial point is quantification over the actual bundled compact group and the actual infinite kernel. No finite carrier appears in Theorem 3.2.

4 Sharp fundamental mode

Define $r_n(t) = \exp(-tn(n+2)/4)$. The vacuum mode is $r_0(t) = 1$. For $n \geq 1$ the elementary inequality

$$3 \leq n(n+2)$$

and monotonicity of the real exponential give

$$r_n(t) \leq r_1(t) = e^{-3t/4} \quad (t \geq 0).$$

For $t > 0$, $0 < r_1(t) < 1$. Consequently the fundamental representation is the sharp slowest mode among all non-vacuum character labels. “Sharp” here means that equality is attained at $n = 1$, not that a Peter–Weyl completeness theorem for the completed class-function Hilbert space has been formalized.

Non-vacuity is exact. The satellite proves $\chi_n(1) = n + 1$; hence $\chi_1(1) = 2$, and function extensionality shows χ_1 cannot be the zero function. The specialization of Theorem 3.2 is therefore a genuine eigenmode:

$$(K_t * \chi_1)(g) = e^{-3t/4} \chi_1(g), \quad \chi_1 \neq 0. \tag{2}$$

5 The exact two-mode operator intertwining

Let $E = \mathbb{R}^2$ and write $a = (a_0, a_1)$. Define the common diagonal step

$$D_t a = (a_0, e^{-3t/4} a_1).$$

The continuous lift and finite lift are

$$\begin{aligned} (\Phi_{\infty} a)(g) &= a_0 \chi_0(g) + a_1 \chi_1(g), \\ (\Phi_2 a)(x) &= a_0 + a_1 \sigma(x), \end{aligned} \quad \sigma(0) = 1, \quad \sigma(1) = -1.$$

These are not informal identifications. The Lean development proves all of the following identities.

Theorem 5.1 (exact two-mode intertwining). *For every $t > 0$:*

(a) Φ_∞ is injective and

$$\int_{\text{SU}(2)} (\Phi_\infty a)(g)(\Phi_\infty b)(g) d\mu_H(g) = a_0 b_0 + a_1 b_1;$$

(b) Φ_2 is bijective onto the entire Euclidean space of the finite carrier, with inverse coefficients

$$a_0 = \frac{v(0) + v(1)}{2}, \quad a_1 = \frac{v(0) - v(1)}{2};$$

(c) actual normalized-Haar heat convolution obeys $H_t \Phi_\infty = \Phi_\infty D_t$; and

(d) the actual finite transfer obeys $W_t \Phi_2 = \Phi_2 D_t$.

Consequently the induced map $Q(\Phi_\infty a) = \Phi_2 a$ is well defined and bijective from the continuous two-mode sector to the finite space, and it satisfies

$$QH_t = W_t Q$$

on that sector.

The key point is operator-level, not scalar-level. The first commuting identity is proved by establishing integrability of the infinite heat kernel against each translated character, distributing the Haar integral, and using the already proved character eigenidentities. The second is proved entrywise from the two carrier sums and the exact entries of W_t . The common map Q is the map induced by two Lean-proved intertwiners; no discretization or sampling assertion is inserted as a hypothesis.

Injectivity of Φ_∞ also has a concrete independent check. For the bundled phase holonomy $p = \text{diag}(i, -i)$, the fundamental character satisfies $\chi_1(p) = 0$, while $\chi_1(1) = 2$. Evaluation at p recovers a_0 , and evaluation at the identity then recovers a_1 . On the finite side $\Phi_2(0, 1)$ is definitionally the nonzero sign vector used in the projected-transfer witness.

6 Finite SU(2)-inhabited transport witness

The finite endpoint is a quotient/interface object, not an approximation in which group axioms are discarded. Its two labels carry two explicit matrices: the identity and a noncentral diagonal phase with determinant one. Both are bundled as elements of SU(2), and their inequality is proved exactly.

For a rate $0 < r < 1$, the transition kernel is

$$W_r = \frac{1}{2} \begin{pmatrix} 1+r & 1-r \\ 1-r & 1+r \end{pmatrix}.$$

Its rows sum to one, it is symmetric and strictly positive, and

$$W_r^k = W_{r^k}.$$

For any observable f on the two carrier points, the band covariance is the identity

$$\text{Cov}_k(f, f) = \left(\frac{f(0) - f(1)}{2} \right)^2 r^k. \quad (3)$$

Thus the decay premise consumed by the Dobrushin theorem is derived, not assumed. The sign observable $(1, -1)$ is orthogonal to the normalized constant vacuum, and the projected transfer sends its first component to r . Since $r > 0$, the projected operator is not zero.

The frozen witness instantiates r by

$$\text{heatRate}(t) = e^{-3t/4}.$$

The new bridge proves this is definitionally the same real number as $r_1(t)$ in the continuous heat theorem. It then combines Equation (2) with the existing endpoint:

Theorem 6.1 (continuous-to-finite transport bridge). *For $t > 0$:*

- (a) $K_t * \chi_1 = \text{heatRate}(t)\chi_1$ *pointwise on $\text{SU}(2)$;*
- (b) $\chi_1 \neq 0$;
- (c) *there exists $m > 0$ for which the norm of the vacuum-projected finite transfer operator is at most e^{-m} ; and*
- (d) *that projected operator is nonzero.*

Theorem 6.1 is retained because it is the published parent gap endpoint. The new content of Theorem 5.1 is strictly stronger than its former scalar bridge: it identifies the complete finite operator with the restriction of the continuous heat operator after transport through Q . It still does not identify the two-state matrix with the full operator on completed $L^2(\text{SU}(2))$; the equality is exactly on the invariant sector $\text{span}_{\mathbb{R}}\{\chi_0, \chi_1\}$.

7 Formal proof architecture

The formal dependency graph is intentionally short:

$$\begin{array}{c}
 \text{satellite character convolution} + \text{heat majorant} \\
 \Downarrow \\
 \text{heatKernelPartial_character_eigen_of_lt} \\
 \Downarrow \text{dominated convergence} \\
 \text{heatKernel_character_eigen} \\
 \Downarrow \text{Casimir ordering} + \chi_1(1) = 2 \\
 \text{heatKernel_fundamental_eigen} \\
 \Downarrow \text{integrability} + \text{linearity} + \text{two explicit lifts} \\
 \text{heatOperator_continuousLift} \quad \text{finiteOperator_finiteLift} \\
 \Downarrow \\
 \text{exact_two_mode_intertwining}.
 \end{array}$$

Table 2 performs the hypothesis-discharge audit requested for the endpoint.

The Lean proof uses classical choice and standard analytic facts inherited from Mathlib, as expected for Haar integration and infinite series. The audit criterion is therefore not an empty `#print axioms` list, but absence of `sorryAx` and absence of project-local axioms. The exact printed axiom sets are captured in the verification log.

8 Reproducibility and frozen revisions

The artifact uses Lean 4 through the parent repository’s pinned toolchain. The satellite is fetched as an exact Lake dependency, not as an unversioned copy. At fabrication time the pins were:

Required fact	Source	Discharge
$t > 0$	bridge input	used for summability and $r < 1$
Haar integrability	compact continuity	satellite continuity lemmas and probability measure
dominating function	heat majorant	summable constant times $m + 1$
character orthogonality	satellite theorem	used termwise before the limit
Casimir ordering	new arithmetic lemma	proved by casting $1 \leq n$ and nonlinear arithmetic
$\chi_1 \neq 0$	new exact lemma	evaluation at the identity gives 2
continuous lift injective	new exact lemma	evaluation at the phase and identity; independently certified by Haar pairing
continuous invariance	new intertwining	dominated integrability, integral linearity, and the two character eigenidentities
finite conjugacy	new intertwining	bijective coefficient lift and exact entrywise matrix algebra
finite kernel laws	frozen witness	explicit two-by-two algebra
decay premise	frozen witness	equality (3), not a hypothesis
projected nonzero	frozen witness	exact sign-vector evaluation

Table 2: Every load-bearing hypothesis and where it is discharged.

Object	Revision
frozen predecessor	59c42d93ec5f4bf660792163c41407d9b1f3d8b4
lean-2d-yang-mills	05c4ec316cb9aa295416670a2578b1c2e77e1c36
Mathlib	07642720480157414db592fa85b626dafb71355b

The decisive commands are:

```
lake update
lake build YangMills.OS.SU2HeatTransport
lake build YangMills.OS.SU2HeatIntertwining
lake env lean Task48Axioms.lean
```

They were run in a fresh Google Colab Pro+ CPU/high-RAM runtime with no GPU. Local desktop work was limited to source editing, Git, hashing, TeX compilation and PDF rendering. A source file on disk was not counted as verified until its `.olean` had been materialized remotely.

9 Scope, limitations, and interpretation

The results support the following exact statement: the infinite Haar heat operator has an invariant Haar-orthonormal real sector spanned by χ_0, χ_1 ; its restriction is conjugate, through an explicit coefficient map, to the entire genuine $SU(2)$ -inhabited two-state transfer operator; and the same finite witness fires the parent projected-gap endpoint non-vacuously.

They do not establish any of the following:

- four-dimensional continuum Yang–Mills or the Clay mass-gap problem;
- a thermodynamic or continuum limit of the finite witness;
- Peter–Weyl completeness, the spectrum on every isotypic class, or an operator-norm computation on the completed full Haar L^2 class space;

- a new proof of the satellite’s Migdal recursion, semigroup law, or exact finite-cellulation area law; or
- that two carrier points exhaust, discretize, or approximate the compact group.

The right interpretation is an exact finite-dimensional invariant restriction. It replaces a previously rate-labelled endpoint by an explicit, injective continuous lift, a bijective finite lift, and two commuting operator identities. This is the minimal operator statement that answers the endpoint question; completion beyond this sector remains separate work.

10 Relation to the literature

The use of heat kernels and character expansions in two-dimensional Yang–Mills goes back to Migdal’s recursion equations [1]. Driver constructed the Yang–Mills measure and related it to the heat equation on compact groups [2]; rigorous compact-surface developments include Sengupta [3] and Lévy [4]. The present paper does not reprove those broad analytic constructions. It contributes a small, exact, inspectable theorem interface inside Lean.

The formalization builds on Mathlib [5] and on the pinned two-dimensional satellite [6]. The repository citation is a commit permalink: it identifies the precise source used by Lake rather than a moving branch.

11 Conclusion

The endpoint is now connected at operator level. The continuous side is an actual Haar integral on bundled $SU(2)$, the infinite series is handled by dominated convergence, and the two-mode lift preserves its exact Haar pairing. The finite side consists of distinct genuine holonomies; its coefficient lift is bijective, its covariance identity is exact, and its projected transfer is nonzero. Both operators commute with the same diagonal map, so $e^{-3t/4}$ is no longer merely a shared label: it is one entry of an exact Lean-checked conjugacy on the specified invariant sector.

References

- [1] A. A. Migdal, “Recursion equations in gauge field theories,” *Soviet Physics JETP* 42 (1975), 413–418. <http://www.jetp.ras.ru/cgi-bin/e/index/e/42/3/p413?a=list>
- [2] B. K. Driver, “YM₂: continuum expectations, lattice convergence, and lassos,” *Communications in Mathematical Physics* 123 (1989), 575–616. doi:10.1007/BF01238812.
- [3] A. Sengupta, *Gauge Theory on Compact Surfaces*, Memoirs of the American Mathematical Society 126, no. 600 (1997). doi:10.1090/memo/0600.
- [4] T. Lévy, *Yang–Mills Measure on Compact Surfaces*, Memoirs of the American Mathematical Society 166, no. 790 (2003). doi:10.1090/memo/0790.
- [5] The mathlib Community, “The Lean Mathematical Library,” in *Proceedings of CPP 2020*, 367–381. doi:10.1145/3372885.3373824.
- [6] L. Eriksson, *lean-2d-yang-mills*, commit 05c4ec316cb9aa295416670a2578b1c2e77e1c36 (2026).